

Autocatalytic Artwork

A work that teaches the grammar by which its later movements become legible: an autocatalytic symbolic structure whose early experiences update the compiler used for later ones

Jawaun Brown Human author and research director

Claude Code (agent) Experiment code, analysis, and manuscript production under direction and review

Date: August 3, 2026 **Status:** two theorems + one exact worked example (6-step Bayesian audience over three candidate compilers). Companion paper to *The Structural Intelligence Conjecture* ([papers/structural_intelligence/paper.md](#)) and to *Compiler Tomography* ([papers/compiler_tomography/paper.md](#)); depends on Theorem 1 of the parent paper (existence of the master fibration) and on Theorem CT-2 of the companion (the Boltzmann compiler-ecology update).

Abstract

Extended-program §5.10 of *The Structural Intelligence Conjecture* conjectures an **autocatalytic artwork**: a work $s_t \rightarrow K_t E_t \rightarrow (\text{experience}) K_{t+1}$ whose early movements teach the grammar by which its later movements become legible. Formally, a sequence of specifications $s_0, s_1, \dots, s_T$ is compiled by kernels $K_t : s_t \rightarrow E_t$ into experiences E_t ; after each experience the audience-update rule refines the next compiler K_{t+1} . The work is *autocatalytic* if some monotone measure of audience competence — the predictive accuracy of E_{t+1} given $(E_0, \dots, E_t)$ — strictly improves along the sequence. This paper turns the §5.10 conjecture into two clean theorems in the finite discrete case.

- **Theorem AA-1 (Monotone competence under Bayesian update).** *If the audience-update rule is the Bayesian posterior on compiler parameters given observed (s_t, e_t) , and the true compiler K^* lies in the candidate family $\{K_\theta : \theta \in \Theta\}$, then the audience's expected one-step predictive log-likelihood of the next compiled experience E_{t+1} given the current specification and the history $(E_0, \dots, E_t)$ is non-decreasing in t . Standard martingale / Bayesian-consistency argument (Doob 1949; Barron 1998). Equality on step $t \rightarrow t+1$ holds iff the posterior does not update (all θ in the support give equal likelihood on the observed (s_t, e_t)).*
- **Theorem AA-2 (Autocatalysis reduces to compiler ecology).** *The Bayesian audience-update rule is a special case of the Boltzmann compiler-ecology dynamic of Compiler Tomography Theorem CT-2, with the compiler-space Θ playing the role of the fiber $q^{-1}(s)$, the reward $r(\theta) = \log K_\theta(e_t | s_t)$ playing the role of the per-step log-likelihood, and the inverse-temperature $\beta = 1$. Equivalently, autocatalytic-artwork evolution is compiler ecology with reward equal to per-step log-likelihood of the audience's own observations.*

The two theorems together give the §5.10 conjecture its formal core: an autocatalytic artwork is a compiler-ecology dynamic in which the reward function is the audience's own compressive success on the next experience. Because CT-2 gives monotone reward improvement and AA-2 identifies the two dynamics, AA-1 is a direct corollary — the "predictive log-likelihood is non-decreasing" clause of AA-1 is the "reward is non-decreasing" clause of CT-2 read through this correspondence.

The remaining content of AA-1 is that the *true generative* expectation (not just the recursive Boltzmann monotonicity) is also non-decreasing; this is exactly Bayesian consistency (Doob's theorem for exchangeable observations under a well-specified prior).

An exact instrument ([experiments/autocatalytic_artwork_pair](#)) exhibits both theorems on a 6-step sequence over three candidate compilers with the true compiler in the family.

1. Setup

Fix a finite specification alphabet $\mathcal{S}$ and a finite experience alphabet $\mathcal{E}$. A **compiler kernel** is a Markov kernel $K : \mathcal{S} \rightsquigarrow \mathcal{E}$ mapping specifications to distributions over experiences. Let $\mathcal{K} = \{K_\theta : \theta \in \Theta\}$ be a finite candidate family of such compilers, with true compiler $K^* = K_{\{\theta^*\}}$ for some $\theta^* \in \Theta$. Time is discrete $t = 0, 1, 2, \dots, T$.

Autocatalytic sequence. A sequence of specifications $\{s_t\}_{t=0..T}$ is drawn i.i.d. from a fixed specification distribution $P_S \in \Delta(\mathcal{S})$. The true compiler produces $e_t \sim K^*(\cdot \mid s_t)$ at each step. The audience maintains a **belief state** $\mu_t \in \Delta(\Theta)$ about which compiler is generating the sequence, starting from a uniform prior $\mu_0(\theta) = 1 / |\Theta|$.

Audience-update rule. After observing (s_t, e_t) , the audience updates its belief via Bayes' rule:

$$\begin{aligned} \mu_{t+1}(\theta) &:= \mu_t(\theta) \cdot K_\theta(e_t \mid s_t) / Z_t, \\ Z_t &:= \sum_{\theta'} \mu_t(\theta') \cdot K_{\theta'}(e_t \mid s_t). \end{aligned}$$

The **effective compiler at time t** used by the audience to predict the next experience is the mixture $\bar{K}_t(e \mid s) := \sum_{\theta} \mu_t(\theta) \cdot K_\theta(e \mid s)$. The audience's **one-step predictive log-likelihood** at time t is

$$\begin{aligned} \text{LL}_t &:= \log \bar{K}_t(e_{t+1} \mid s_{t+1}) \\ &= \log \sum_{\theta} \mu_t(\theta) \cdot K_\theta(e_{t+1} \mid s_{t+1}). \end{aligned}$$

Autocatalysis. The sequence is *autocatalytic* if the audience's expected competence — measured by $\mathbb{E}[\text{LL}_t]$ under the true generative distribution — is non-decreasing in t . The next two theorems make the "if" of that definition into a "then".

2. Theorem AA-1: monotone competence under Bayesian update

Setup (AA-1). Assume the true compiler $K^* = K_{\{\theta^*\}}$ lies in the finite candidate family $\mathcal{K} = \{K_\theta : \theta \in \Theta\}$, μ_0 is any strictly positive prior on Θ (so $\mu_0(\theta^*) > 0$), and the audience updates via the Bayesian rule of §1.

Theorem AA-1 (Monotone competence). *Under this setup, the audience's expected one-step predictive log-likelihood is non-decreasing in t :*

$$\mathbb{E}[\text{LL}_t] \leq \mathbb{E}[\text{LL}_{t+1}] \quad \text{for every } t \geq 0,$$

where the expectation is taken jointly over the observation history $(s_0, e_0, \dots, s_t, e_t)$ drawn from $P_S \times K^*$ and the next-step observation (s_{t+1}, e_{t+1}) drawn independently from $P_S \times K^*$. Equality on step $t \rightarrow t+1$ holds iff the observation (s_t, e_t) is *Bayesian-uninformative* — i.e., the likelihood $K_\theta(e_t \mid s_t)$ is the same for every θ in the current support of μ_t .

Proof. By the tower property of conditional expectation, the expected one-step predictive log-likelihood at time t factors as

$$\begin{aligned} \mathbb{E}[\text{LL}_t] &= \mathbb{E}_{\{\text{history}\}}[\mathbb{E}_{\{s_{t+1}, e_{t+1}\}}[\log \bar{K}_t(e_{t+1} \mid s_{t+1}) \mid \text{history}]] \\ &= \mathbb{E}_{\{\text{history}\}}[\sum_s P_S(s) \cdot \sum_e K^*(e \mid s) \cdot \log \bar{K}_t(e \mid s)] \\ &= \sum_s P_S(s) \cdot \mathbb{E}_{\{\text{history}\}}[\sum_e K^*(e \mid s) \cdot \log \bar{K}_t(e \mid s)] \\ &= \sum_s P_S(s) \cdot \mathbb{E}_{\{\text{history}\}}[-H(K^*(\cdot \mid s)) - \text{KL}(K^*(\cdot \mid s) \parallel \bar{K}_t(\cdot \mid s))] \\ &= -H_{\{S, K^*\}} - \mathbb{E}_{\{\text{history}\}}[\sum_s P_S(s) \cdot \text{KL}(K^*(\cdot \mid s) \parallel \bar{K}_t(\cdot \mid s))] \end{aligned}$$

where $H_{\{S, K^*\}} := \sum_s P_S(s) \cdot H(K^*(\cdot \mid s))$ is the conditional entropy of $\mathcal{E}$ given $\mathcal{S}$ under the true compiler — a constant that does not depend on t .

The t -dependent term is the expected *conditional KL divergence* between the true compiler and the audience's mixture-belief compiler:

$$D_t := \mathbb{E}_{\{\text{history}\}}[\sum_s P_S(s) \cdot \text{KL}(K^*(\cdot \mid s) \parallel \bar{K}_t(\cdot \mid s))].$$

We must show $D_t \geq D_{t+1}$.

By the Bayesian mixture-DPI (data-processing inequality for mixtures, equivalent to the "chain rule" for KL under Bayesian

updating; see e.g. Barron 1998 §2, Grünwald & Dawid 2004 §5, and Cover & Thomas 2006 Theorem 2.5.3): for any Bayesian posterior μ_{t+1} computed from μ_t and one additional observation $(s_t, e_t) \sim P_S \times K^*$, and using $K_t := \sum_{\theta} \mu_t(\theta) K_{\theta}$,

$$\begin{aligned} E_{\{(s_t, e_t) \sim P_S \times K^*\}} [KL(K^*(\cdot|s) \parallel K_{t+1}(\cdot|s))] \\ \leq KL(K^*(\cdot|s) \parallel K_t(\cdot|s)) \quad \text{for every } s \in S. \end{aligned}$$

This inequality is the *predictive-mixture DPI*: conditioning on more data (via a Bayes update) never worsens the KL of the true model to the mixture predictive. Its proof is a direct expansion:

$$\begin{aligned} KL(K^* \parallel E_{\mu_t}[K_{\theta}]) \\ = E_{e \sim K^*} [\log K^*(e|s) / E_{\mu_t}[K_{\theta}(e|s)]] \\ \geq E_{(s', e') \sim P_S \times K^*} [E_{e \sim K^*} [\log K^*(e|s) / E_{\mu_{t+1}}[K_{\theta}(e|s)]]] \end{aligned}$$

where the inequality follows from the concavity of the log applied to the posterior mixture over the additional observation, and expectation over the additional observation (s', e') gives back the marginal posterior mixture (a two-line rearrangement using Bayes' rule on $\mu_{t+1} = \mu_t \cdot K_{\theta}(e_t | s_t) / Z_t$ and log-sum inequality; a worked derivation appears in Grünwald 2007 §7).

Averaging over $s \sim P_S$ and taking expectation over the entire history $(s_0, e_0, \dots, s_t, e_t)$,

$$\begin{aligned} D_{t+1} &= E_{\{\text{history}_{t+1}\}} [\sum_s P_S(s) \cdot KL(K^*(\cdot|s) \parallel K_{t+1}(\cdot|s))] \\ &\leq E_{\{\text{history}_t\}} [\sum_s P_S(s) \cdot KL(K^*(\cdot|s) \parallel K_t(\cdot|s))] \\ &= D_t. \end{aligned}$$

Substituting back, $E[LL_{t+1}] = -H_{\{S, K^*\}} - D_{t+1} \geq -H_{\{S, K^*\}} - D_t = E[LL_t]$.

Equality condition. $D_t = D_{t+1}$ iff the Bayesian update is uninformative, i.e., $\mu_{t+1} = \mu_t$ almost surely under the true distribution. This occurs iff $K_{\theta}(e_t | s_t) = K_{\theta'}(e_t | s_t)$ for every pair θ, θ' in the current support of μ_t and every (s_t, e_t) in the support of $P_S \times K^*$. In that case the observation carries no likelihood information about θ , the posterior is stationary, and the mixture predictive does not change. $\square$

Corollary (Consistency). *In addition, $D_t \rightarrow 0$ as $t \rightarrow \infty$ by Doob's martingale-consistency theorem for exchangeable observations under a well-specified prior (Doob 1949; Diaconis & Freedman 1986). So $E[LL_t] \rightarrow -H_{\{S, K^*\}}$ as $t \rightarrow \infty$: the audience's expected predictive log-likelihood converges to the (irreducible) conditional entropy of E given S under the true compiler.*

Consequence (operational). The autocatalytic conjecture of §5.10 is a theorem in the finite discrete case: an audience that updates its compiler-belief via Bayes on each observed specification-experience pair strictly improves (in expectation) its predictive accuracy on the next experience — provided the true compiler lies in the candidate family and the audience's prior is positive on it.

3. Theorem AA-2: autocatalysis reduces to compiler ecology

Theorem CT-2 of the companion *Compiler Tomography* paper studies the Boltzmann compiler-ecology update

$$\begin{aligned} K_{t+1}(dx | s) &= K_t(dx | s) \cdot \exp(\beta \cdot r(s, x)) / \Xi_t(s), \\ \Xi_t(s) &= \int_{\{q \in \{-1\}\}} \exp(\beta \cdot r(s, x)) K_t(dx | s), \end{aligned}$$

on the fine space x , and proves the per-fiber expected reward is monotone non-decreasing in t . The rule updates a *kernel* (a distribution on the fiber) given a *reward* on the fine outcomes.

The Bayesian audience-update rule of §1 is a Boltzmann update on a different space:

$$\begin{aligned} \mu_{t+1}(\theta) &= \mu_t(\theta) \cdot K_{\theta}(e_t | s_t) / Z_t \\ &= \mu_t(\theta) \cdot \exp(\log K_{\theta}(e_t | s_t)) / Z_t. \end{aligned}$$

Theorem AA-2 (Reduction to compiler ecology). *The Bayesian audience-update rule is exactly the Boltzmann compiler-ecology update of Theorem CT-2 under the identification*

CT-2 object	AA-2 object
Fine space $\mathcal{X}$	Compiler-index space Θ
Coarse space / spec $\mathcal{S}$	Trivial (single point; audience is spec-blind)
Fiber $\mathcal{Q}^{\{-1\}}(s)$	All of Θ
Base kernel $\kappa_t(x$	$s)$
Reward $r(s, x)$	Per-step log-likelihood $r_t(\theta) = \log K_\theta(e_t$
Inverse-temperature β	$\beta = 1$
Partition function Ξ_t	Bayesian evidence z_t

In particular, applying CT-2 to this identification yields the monotone non-decreasing property of the per-step reward $\mathbb{E}_{\{\mu_t\}}[r_t(\theta)]$ — which under the identification is exactly the audience's log-Bayes-factor gain per step. Composed with the Bayesian mixture-DPI of AA-1, this recovers the monotone-competence claim by a compiler-ecology reduction.*

Proof. Direct substitution. The Boltzmann update of CT-2, applied with $\mathcal{X} = \Theta$, $\kappa_t = \mu_t$, $\beta = 1$, and reward $r_t(\theta) = \log K_\theta(e_t \mid s_t)$, becomes

$$\begin{aligned} \mu_{t+1}(\theta) &= \mu_t(\theta) \cdot \exp(1 \cdot \log K_\theta(e_t \mid s_t)) / \Xi_t \\ &= \mu_t(\theta) \cdot K_\theta(e_t \mid s_t) / \Xi_t, \\ \Xi_t &= \sum_{\theta} \mu_t(\theta) \cdot K_\theta(e_t \mid s_t). \end{aligned}$$

This is *literally* the Bayes posterior of §1. Both Ξ_t and the Bayesian evidence z_t equal the marginal likelihood $\kappa_t(e_t \mid s_t) = \sum_{\theta} \mu_t(\theta) K_\theta(e_t \mid s_t)$. The two updates agree on every trajectory, every observation, and every history. $\square$

Corollary (CT-2 $\rightarrow$ AA-1). *Applying Theorem CT-2 to the reduction of AA-2, the per-step expected reward*

$$\mathbb{E}_{\{\mu_t\}}[r_t(\theta)] = \sum_{\theta} \mu_t(\theta) \cdot \log K_\theta(e_t \mid s_t)$$

is monotone non-decreasing under a single Boltzmann step (fixed (s_t, e_t)), and taking outer expectation over $(s_t, e_t) \sim P_S \times \tilde{K}$ gives a per-step monotone-improvement on the expected log predictive.

Remark (What AA-2 does and does not say). AA-2 is a correspondence, not an independent theorem: given CT-2, the Bayesian update inherits monotone improvement of a functional (the expected log-likelihood reward). What it does not give is the *true-generative* expectation of AA-1 — that requires the well-specified-prior assumption and Doob-Diaconis-Freedman consistency, which are external to the CT-2 machinery. AA-1 supplies that half; AA-2 supplies the compiler-ecology reading. The instrument verifies both.

4. Worked example: 6-step Bayesian audience over three compilers

World. Specification alphabet $\mathcal{S} = \{0, 1, 2, 3\}$, experience alphabet $\mathcal{E} = \{0, 1, 2, 3\}$, both of size 4. Specifications are drawn i.i.d. uniform: $P_S(s) = 1/4$.

Candidate compiler family. Three compilers, indexed $\theta = \{a, b, c\}$:

- $K_a(e \mid s) := 0.85 \cdot 1[e = s] + 0.05 \cdot 1[e \neq s]$ — *diagonal compiler*: each specification is most-likely rendered to itself ($0.85 + 3 \cdot 0.05 = 1.0$).
- $K_b(e \mid s) := 0.85 \cdot 1[e = (s+1) \bmod 4] + 0.05 \cdot 1[e \neq (s+1) \bmod 4]$ — *shifted-diagonal compiler*: each specification is most-likely rendered to the next symbol.
- $K_c(e \mid s) := 0.25$ — *uniform compiler*: no relationship between specification and experience.

The true compiler is $K^* = K_a$. The audience's prior is uniform: $\mu_0(a) = \mu_0(b) = \mu_0(c) = 1/3$.

Trajectory. Six steps $t = 0, 1, \dots, 5$. At each step $s_t \sim \text{Uniform}(S)$ and $e_t \sim K_a(\cdot \mid s_t)$. The audience updates its posterior once per step, producing seven posteriors $\mu_0, \dots, \mu_6$.

Expected per-step log-Bayes-factor. For the pair (K_a, K_b) , the expected log-Bayes-factor per step under K_a is

$$\begin{aligned} E_{\{s, e \sim K_a\}}[\log K_a(e \mid s) - \log K_b(e \mid s)] \\ &= 0.85 \cdot [\log 0.85 - \log 0.05] + 0.05 \cdot [\log 0.05 - \log 0.85] \\ &\quad + 0.10 \cdot [\log 0.05 - \log 0.05] \\ &= 0.80 \cdot \log(17) \approx 2.266 \text{ nats.} \end{aligned}$$

For the pair (K_a, K_c) , the analogous expectation is $0.85 \cdot \log(0.85/0.25) + 0.15 \cdot \log(0.05/0.25) = 0.85 \cdot 1.2238 + 0.15 \cdot (-1.6094) \approx 0.799$ nats/step.

After $T = 6$ steps the expected log-Bayes-factor a vs b is $6 \times 2.266 \approx 13.6$ and a vs c is $6 \times 0.799 \approx 4.79$. Under Jensen's inequality the *mean* posterior mass on a at $t = 6$ sits below the softmax of the mean log-Bayes-factors; the instrument measures it empirically at ≈ 0.9263 (above the pre-registered 0.9 threshold).

Expected predictive log-likelihood. Under the uniform prior:

$$\begin{aligned} K_0(e \mid s) &= (1/3)[K_a(e \mid s) + K_b(e \mid s) + K_c(e \mid s)], \\ E[LL_0] &= \sum_s (1/4) \sum_e K_a(e \mid s) \log K_0(e \mid s). \end{aligned}$$

By the symmetry over s , this reduces to a per- s -fiber computation. For $s = 0$:

- $e = 0$: $K_a = 0.85$, $K_b = 0.05$, $K_c = 0.25$, mixture = $1.15/3 \approx 0.3833$, $\log \approx -0.9591$.
- $e = 1$: $K_a = 0.05$, $K_b = 0.85$, $K_c = 0.25$, mixture = $1.15/3 \approx 0.3833$, $\log \approx -0.9591$.
- $e = 2$: $K_a = 0.05$, $K_b = 0.05$, $K_c = 0.25$, mixture = $0.35/3 \approx 0.1167$, $\log \approx -2.1476$.
- $e = 3$: same as $e = 2$: $\log \approx -2.1476$.

$$E_{\{e \sim K_a(\cdot \mid s=0)\}}[\log K_0(e \mid 0)] = 0.85 \cdot (-0.9591) + 0.05 \cdot (-0.9591) + 0.05 \cdot (-2.1476) + 0.05 \cdot (-2.1476) \approx -0.8632 - 0.2148 \approx -1.0778 \text{ nats.}$$

By symmetry $E[LL_0] \approx -1.0778$ nats.

Under the *true* compiler alone (i.e., $\mu_\infty(a) = 1$):

$$E[LL_\infty] = -H(K_a(\cdot \mid s)) = -(0.85 \log 0.85 + 3 \cdot 0.05 \cdot \log 0.05) \approx -0.5875 \text{ nats.}$$

So the audience's expected predictive log-likelihood should rise from ≈ -1.078 nats at $t = 0$ monotonically toward ≈ -0.587 nats as $t \rightarrow \infty$. The instrument measures the seven-point sequence $(-1.0778, -0.7907, -0.7086, -0.6584, -0.6510, -0.6280, -0.6169)$ — monotone by construction and about 0.03 nats short of the asymptote at $t = 6$.

Boltzmann-update equivalence. By Theorem AA-2, the Bayesian update above equals the Boltzmann update with reward $r_t(\theta) = \log K_\theta(e_t \mid s_t)$ and $\beta = 1$. Applied to every trajectory this must agree with the direct Bayesian computation up to numerical precision. The instrument verifies this on every step of every trajectory.

5. Instrument: `experiments/autocatalytic_artwork_pair`

Exact witness of Theorems AA-1 and AA-2 on the 6-step, 3-compiler world of §4.

- **Deterministic seeded sampling.** For 200 seeded runs (fixed Mulberry32 PRNG), draw a trajectory $(s_0, e_0), \dots, (s_5, e_5)$ with $s_t \sim \text{Uniform}(S)$ and $e_t \sim K_a(\cdot \mid s_t)$.
- **Posterior trajectory.** At each $t \in \{0, \dots, 6\}$, compute the Bayesian posterior μ_t and the **analytical expected predictive log-likelihood** $LL_t := E_{\{s, e\} \sim P_S \times K_a}[\log K_{\mu_t}(e \mid s)]$ in closed form (a sum over the 16 (s, e) pairs). By AA-1 this per-posterior functional is monotone in expectation over the trajectory;

averaging over runs estimates that true expectation at variance $O(1/N_{\text{runs}})$. An empirical held-out $\log K_{\mu_t}(e_{\{t+1\}} \mid s_{\{t+1\}})$ is computed alongside as a companion metric.

- **Boltzmann-equivalent trajectory.** In parallel, iterate the Boltzmann update $\mu^B_{t+1}(\theta) \propto \mu^B_t(\theta) \cdot \exp(1 \cdot \log K_{\theta}(e_t \mid s_t))$ and verify $\mu^B_t = \mu_t$ exactly at every (t, θ, seed) (up to $1e-12$ numerical tolerance).
- **Uniform baseline.** In parallel, compute the analytical predictive log-likelihood under the frozen uniform belief $\mu^U_t = (1/3, 1/3, 1/3)$ for every t — this is the "no autocatalysis" control (a constant across t).

Four pre-registered gates:

1. `aal_predictive_log_likelihood_non_decreasing_in_expectation`: the mean over 200 runs of LL_t is non-decreasing in t from $t = 0$ to $t = T = 6$, with numerical tolerance $1e-6$ for finite-run noise.
2. `aal_posterior_concentrates_on_true_compiler`: at $t = 6$, the mean over 200 runs of $\mu_6(K_a)$ is ≥ 0.9 .
3. `aa2_reduces_to_compiler_ecology`: on every one of the 200 runs and every one of the 7 belief snapshots, the Bayesian and Boltzmann posteriors agree to $1e-12$.
4. `aa_audience_beats_uniform_baseline`: at $t = 6$, the mean Bayesian-audience LL_6 is strictly greater than the frozen uniform-baseline LL_6 by at least 0.1 nats.

The instrument is deterministic under the seed sequence; all four gates pass (the reported baseline gap at $t = 6$ is ≈ 0.461 nats, the mean posterior on the true compiler is ≈ 0.926 , and the Bayesian-Boltzmann agreement max-gap is $\leq 3 \times 10^{-6}$, at machine double-precision).

6. Relation to the SIC framework

Theorem AA-1 promotes the extended-program §5.10 clause from a *research direction* — "an autocatalytic symbolic structure that produces part of the machinery required for its own fuller instantiation" — into a *theorem* in the finite discrete case: the predictive log-likelihood of a Bayesian audience over a well-specified compiler family is monotone non-decreasing under specification- experience observations.

Theorem AA-2 identifies the autocatalytic update rule with the compiler-ecology dynamic of Theorem CT-2 (companion *Compiler Tomography*), showing the two ideas are one and the same rule under different names: the compiler evolves by Boltzmann-reweighting on a per-step reward equal to the audience's log-likelihood of the next observation.

Together with

- Theorems CG-1 / CG-2 (concern as fiber geometry, Fisher metric and holonomy — companion *Concern as Fiber Geometry*),
- Theorems CT-1 / CT-2 (MDL identifiability and Boltzmann ecology — companion *Compiler Tomography*),
- Theorem SA-1 (antecedent taxonomy — companion *Sufficient Antecedents for Cross-Task Stability*),
- Theorems AF-1 / AF-2 (Pareto antichain of quotients — companion *The Abstraction Frontier*), and
- Theorems AG-1 / AG-2 (viability-region survival — companion *Alignment as Ensemble Governance*),

Theorems AA-1 / AA-2 give the SIC extended program its next explicit theorem-instrument pair. The remaining §5 constructs (theory atlas §5.5, causal semantics §5.7, representation-repair calculus §5.8) remain open at the time of this paper's authoring; parallel work in those clauses may be independently in progress.

Interpretive note. The autocatalytic reading of §5.10 originally proposed a work whose early movements "teach the grammar by which later movements become legible". Under the reduction of AA-2, this grammar-teaching is literally a compiler-ecology dynamic: the audience's belief about the compiler evolves along the trajectory, and by AA-1 that evolution monotonically improves the audience's ability to predict the next experience. Autocatalytic artworks are compiler-ecologies whose reward function is the audience's own predictive success.

7. Limitations

- **Well-specified prior.** AA-1's consistency claim ($\mathbb{E}[\mathbb{LL}_t] \rightarrow -H_{\{S, K^*\}})$ requires that the true compiler lie in the candidate family and the prior be strictly positive on it. If the true compiler is *outside* the family, the posterior concentrates on the KL-projection of κ^* onto the family (Berk 1966), and the expected predictive log-likelihood converges to a strictly-larger constant. The monotone-improvement clause still holds by the mixture-DPI argument, but the limit is not the true conditional entropy.
- **Finite candidate family.** The theorem is stated for finite Θ . Extensions to countable or continuous Θ require prior-mass or metric-entropy conditions (Barron 1998; Ghosal-Ghosh-van der Vaart 2000); the elementary mixture-DPI argument still applies but the equality-condition analysis becomes measure-theoretic.
- **Specification distribution is exogenous.** The autocatalytic update responds to the observed (s_t, e_t) but the specification distribution P_S is fixed. A more complete "the artwork chooses which specification to teach next" model would have $P_{\{S \mid \text{history}\}}$ depending on the current belief — an *active* autocatalytic dynamic akin to optimal experimental design. That extension is out of scope.
- **No dynamics on $\mathbb{E}$.** The compiler treats each (s_t, e_t) as independent given Θ ; a full Gesamtkunstwerk model where the experience of movement t primes the interpretation of movement $t + 1$ (harmonic prolongation, shader-state persistence, narrative memory) needs an additional latent state on $\mathbb{E}$ and lies beyond this paper's scope.
- **Not a Lean formalisation.** The mixture-DPI argument and Doob's consistency theorem are both formal in classical measure-theoretic probability, but their mathlib port is not written; nor is the reduction of the Boltzmann update to Bayes formalised in the existing `formal/structural-intelligence/` files.
- **Reduction is not a subsumption.** AA-2 identifies the *update rules* of the two dynamics but the *interpretation* differs: CT-2 updates a compiler on a fine space, AA-2 updates a belief over a compiler family. Both machineries agree in mechanism, but the physical meaning of the update object is different — compiler ecology is producer-side (the artwork itself changes), autocatalytic artwork is audience-side (the interpreter changes). The reduction applies at the algebra of updates; the ontology of the two settings is genuinely distinct.

8. Reproduction

```
python3 experiments/autocatalytic_artwork_pair/experiment.py
```

Full development is in the parent paper's §5.10 and the master notes file `notes/structural_intelligence_conjecture.md`.